

HR Attrition Insights & Process Improvement

Requirements Document & KPI Catalog · Simarjeet Singh

1. Business Context

Voluntary attrition is one of the largest controllable cost drivers in any people-heavy organization. Replacing a mid-level employee typically costs 50–150% of annual salary in recruitment, onboarding, and lost productivity. The HR leadership team needs a single, trusted view of **who is leaving, why they leave, and where the highest-risk pockets sit**, so retention budgets and process improvements can be targeted, not sprayed across the workforce.

2. Stakeholders

Stakeholder	Role	Primary need
CHRO	Sponsor	Org-level attrition trend, cost exposure, ROI of retention spend
HRBPs	Primary user	Department / role-level drivers, action lists for managers
Talent Acquisition	Consumer	Forecast of replacement hiring demand
Finance	Consumer	Replacement cost projection by quarter
People Managers	Consumer	Team-level dashboard, flight-risk indicators

3. Business Questions to Answer

- What is the overall attrition rate, and how does it trend by department, role, and tenure band?
- Which job roles and compensation bands drive a disproportionate share of leavers?
- How much does overtime, work-life balance, and job satisfaction influence attrition?
- Which employees are flight risks today (under-paid relative to role + overtime + low satisfaction)?
- What is the projected replacement cost if current attrition continues for the next 4 quarters?

4. Functional Requirements

- Single Power BI report (.pbix) refreshed daily from the HRIS data warehouse.
- Filters: Date, Department, Job Role, Tenure Bucket, Compensation Band, OverTime, Gender.
- Pages: (1) Executive Summary, (2) Driver Deep Dive, (3) Flight-Risk Cohort, (4) Cost Impact.
- Drill-through from any role/department to an employee-level table (PII-restricted by RLS).
- Export-to-Excel disabled for the employee-level page (data privacy).

5. Non-Functional Requirements

- Row-Level Security: HRBPs only see their assigned business units; CHRO sees all.
- Refresh SLA: 06:00 local, < 10 min total refresh duration.
- Data retention: 36 months of monthly snapshots for trend analysis.
- Accessibility: contrast AA, no color-only encoding of attrition.

6. KPI Catalog

Each KPI lists the business definition, the DAX-style formula, and the owner.

KPI	Definition	Formula (DAX)	Owner
Attrition Rate %	Leavers in period ÷ avg headcount	DIVIDE([Leavers], [Avg Headcount])	CHRO
Voluntary Leavers	Count of employees with Attrition='Yes' and reason=Voluntary	CALCULATE(COUNTROWS(Employees), Employees[Attrition]="Yes")	HRBP
Avg Tenure (yrs)	Average YearsAtCompany of active employees	AVERAGE(Employees[YearsAtCompany])	HRBP
Early Attrition %	% of leavers with tenure < 2 years	DIVIDE(CALCULATE([Leavers], Employees[YearsAtCompany]<2), [Leavers])	TA Lead
OverTime Attrition Lift	Attrition rate among OT=Yes minus OT=No	[Attrition % OT Yes] - [Attrition % OT No]	HRBP
Comp-Gap Risk Count	Headcount earning <85% of role average and on OT=Yes	Measure built on RANKX/AVERAGEX over JobRole	HRBP
Replacement Cost (\$)	Leavers × (Avg Monthly Income × 6)	[Leavers] * AVERAGE(Employees[MonthlyIncome]) * 6	Finance
Avg Job Satisfaction	Mean of 1–4 satisfaction score	AVERAGE(Employees[JobSatisfaction])	HRBP
Avg Work-Life Balance	Mean of 1–4 WLB score	AVERAGE(Employees[WorkLifeBalance])	HRBP
Promotion Lag (yrs)	Avg YearsSinceLastPromotion among active	AVERAGE(Employees[YearsSinceLastPromotion])	HRBP

7. Recommended Process Improvements

- Stay-interview program for all employees with tenure < 18 months in the top-3 attrition roles.
- Compensation review cycle for any employee earning below 85% of role average.
- Mandatory OT cap (≤ 8 hrs/week) for departments where OT=Yes attrition lift exceeds 20 pts.
- Quarterly Pulse survey feeding JobSatisfaction and WorkLifeBalance scores into the dashboard.
- Manager enablement: every people manager sees the team-level page in their 1:1 cadence.

8. Deliverables

- data/hr_employees.csv — synthetic employee dataset (1,500 rows).
- sql/attrition_analysis.sql — driver queries (PostgreSQL/MySQL/SQL Server compatible).
- python/eda.py + python/eda_charts.pdf — exploratory data analysis.
- excel/HR_Attrition_Dashboard.xlsx — pivots, KPIs, and charts.
- powerbi/measures.dax + powerbi/README.md — DAX measures and dashboard build guide.

- docs/Requirements_and_KPI_Catalog.pdf — this document.